EE4437 Assignment 2026

Please read these instructions carefully

1. **Submission**: Please submit your assignment to Canvas > Assignments > Homework Assignment for Part II. Hard copy submission, e-mail submission, or submission to the wrong assignment/folder will not be considered.

2. **Deadline**: Sunday, 19 April 2026 (11 pm). Late submission will incur a penalty. No marks will be awarded for submissions after the solutions have been released.

3. **Format**: Submit as a single pdf file. If you are using e.g. Microsoft Word, or scanning your solutions as images, convert and combine them into a *single PDF file* before submission.

4. **File Name** format is: [Matric no.]_[Name].pdf, e.g.: A012345B_John_Tan.pdf.

5. **Resubmission**: If you wish to make changes after submission but before the deadline, delete the old submission and submit the modified version. If a student has multiple submissions under his/her name, only the latest version submitted before the deadline will be considered.

6. **No compressed files**: Zipped or compressed files will not be considered.

7. **Answer all questions**. The total mark is 30. This assignment carries a CA weightage of 10%.

8. You may make use of the following:

   Electron charge, $e = 1.6 \times 10^{-19}$ C

   Permittivity of vacuum, $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

   Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J/K

   Thermal energy, $k_B T = 25.9$ meV at temperature of 300 K

   Mass of electron in vacuum, $m_e = 9.1 \times 10^{-31}$ kg

   Planck's constant, $h = 6.63 \times 10^{-34}$ J s

   Reduced Planck's constant, $\hbar = 1.05 \times 10^{-34}$ J s

   $E(eV) = 1.24/\lambda(\mu m)$

Question 1

(a) A boron-doped Si photoconductor has a resistance of 160 Ω in the absence of infra-red (IR) radiation. When IR radiation of 20 mW/m^2 is incident on the photoconductor, the resulting photocurrent constitutes 75% of the total current at steady state. Calculate the resistance of the photoconductor at steady state.

(5 marks)

In the dark, the resistance is $R_0 = 160\ \Omega$.
At $\Phi_p = 20\ \mathrm{mW/m^2}$, the photocurrent constitutes 75% of total current.
Let dark current be I_0.
Therefore,

$$\frac{I_p}{I_p + I_0} = \frac{3}{4} \Rightarrow I_p = 3I_0$$

Resistance is inversely proportional to current. Hence,

$$\frac{R_1}{R_0} = \frac{I_0}{I_0 + I_p} = \frac{1}{1+3} = \frac{1}{4}$$
$$\Rightarrow R_1 = 0.25\,R_0 = 40\ \Omega$$

(b) The photoconductor in Q.1(a) is then exposed to infra-red (IR) radiation of 50 mW/m^2. Calculate its new resistance at steady state.

(5 marks)

Photocurrent I_p is proportional to the optical power Φ_p.
At $\Phi_p = 50\ mW/m^2$, the photocurrent is increased i.e. to $I_{p1} = \frac{50}{20} I_p = \frac{5}{2}(3I_0) = \frac{15}{2} I_0$.

Therefore, corresponding resistance is given by

$$\frac{R_2}{R_0} = \frac{I_0}{I_0 + I_{p1}} = \frac{1}{1+7.5} = \frac{1}{8.5}$$
$$\Rightarrow R_1 = \ R_0/8.5 = 18.8\ \Omega$$

(b) Optical radiation of 70 μW power is incident on an open-circuited pn photodiode. The photodiode attains an open circuit voltage responsivity of 3.6×10^3 V/W at steady state. If the negative saturation current of the photodiode is 1.5 nA and the wavelength of the incident light is 650 nm, calculate the quantum efficiency of the photodiode. You may assume room temperature of 300 K.

(6 marks)

$$V_{oc} = R_{oc} \times \Phi_p = 3.6 \times 10^3 \times 70 \times 10^{-6} = 0.252$$

$$I_p = I_0\left[\exp\left(\frac{qV_{oc}}{kT}\right) - 1\right] = I_0\left[\exp\left(\frac{0.252}{25.9\times10^{-3}}\right) - 1\right] \approx I_0\left[\exp\left(\frac{0.252}{25.9\times10^{-3}}\right)\right]$$
$$= (1.5 \times 10^{-9})\exp\left(\frac{0.252}{25.9 \times 10^{-3}}\right) = 2.52 \times 10^{-5}\mathrm{A}$$

But $I_p = e\eta\left(\frac{\Phi_p}{h\nu}\right) \Rightarrow 2.52 \times 10^{-5}\text{A} = \frac{e\eta(70\times10^{-6})}{\left(\frac{1.24e}{0.65}\right)} \Rightarrow \eta = 0.687.$

Question 2

A quantum well infra-red photodetector (QWIP) consists of alternate layers of semiconductors A and B, the dimensions and material parameters of which are given in the table below.

	A	B
Thickness of each layer	12 nm	20 nm
Band gap energy	0.73 eV	1.86 eV
Effective electron mass	0.08 m_e	0.28 m_e

(N.b.: m_e is the electron mass in vacuum).

If the quantum wells formed in the valence and conduction bands are of equal depth, evaluate the following:

(a) Estimate the wavelength corresponding to the main absorption peak of the QWIP using the infinite quantum well model. State any other assumptions in your calculations.

(6 marks)

$$E_n = \frac{\hbar^2 k^2}{2m^*} = \frac{\hbar^2}{2m^*}\left(\frac{n^2\pi^2}{L^2}\right)$$

The well material is A as it has a smaller bandgap.

The maximum transition occurs at the wavelength corresponding to the transition between quantum levels $n = 1$ to $n = 2$, i.e.,

$$\begin{aligned} \Delta E &= E_2 - E_1 \\ &= \frac{\hbar^2\pi^2}{2mL^2}(4-1) \\ &= 1.56 \times 10^{-21}\text{J} \\ &= 0.0973 \text{ eV} \end{aligned}$$

This corresponds to a wavelength of $\lambda_0 = \frac{1.24}{0.0973} = 12.7\mu\text{m}.$

Other assumption apart from the infinite quantum well assumption: i) only one dimension is small, the other two dimensions being macroscopic (1-dimensional quantum well), (ii) dominant transmission is between $n = 1$ to $n = 2$, other transitions (collectively ~ 4%) can be neglected.

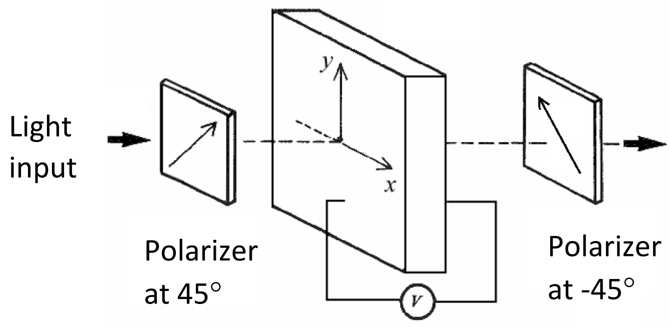
(b) Is your estimated wavelength longer or shorter than the actual wavelength? Explain your answer.

(3 marks)

- The approximation assumes an infinite square quantum well and confinement in one direction. In actual fact the QW's are finite in depth and are confined in all three dimensions.
- The degree of confinement is less than that assumed in a 1-dimensional infinite QW model. The wavefunction leaks into the barrier region. Hence, the effective confinement width is larger than the actual physical well width.
- Therefore, the energy level separation in an actual finite QW is less than that predicted by the infinite QW model.
- Hence, the actual energy separation between the $n = 1$ and $n = 2$ levels is less than that calculated above, i.e., $\Delta E < 0.0973\ \mathrm{eV}$.
- Therefore, the actual peak lies at wavelength $\lambda > \lambda_0 = 12.7\ \mu\mathrm{m}$, i.e., the estimated wavelength is shorter than the actual wavelength.

Question 3

The figure below depicts an electro-optic crystal modulator placed between two parallel polarizers, oriented at ±45° to the *x* and *y* principal axes of the modulator. The refractive index n_0 of the crystal in the absence of electric field is 2.7, while its Pockels coefficient is 2.4×10^{-11} m/V along the *x*-axis and -2.4×10^{-11} m/V along the *y*-axis. The light at the input is originally unpolarized and has a wavelength of λ_0 = 550 nm in vacuum.



Determine the phase difference between the components of light polarized in the *x* and *y* directions after passing through the crystal. Express your answer in terms of *V*, the voltage applied across the modulator.

(5 marks)

$$\Delta n = (n_y - n_x) = -\frac{n_0^3(r_y - r_x)V}{2d} = 4.72 \times 10^{-10}(V/d),$$

where d is the crystal thickness. The phase difference is given by

$$\Delta\phi = \frac{(2\pi d\,\Delta n)}{\lambda_0}$$
$$= \frac{2\pi d}{\lambda_0} \times \frac{n_0^3(r_x - r_y)V}{2d}$$
$$= \frac{\pi n_0^3(r_x - r_y)V}{\lambda_0}$$
$$= \frac{\pi(2.7)^3(4.8 \times 10^{-11})}{550 \times 10^{-9}}V = 5.4 \times 10^{-3}V$$